

TECHWIZ 7

Project Name: Cyberpunk Detective

Theme: Sci-Fi Mystery Game

Category: Unity Game Foundry

Instructor: **Group: 5 JOKES**

Nguyen Hong Son Member:

1. Student1694623-Trinh Tan Loc
2. Student1704619-Tran Quoc An
3. Student1704614-Pham Thien Phuc
4. Student1694630-Nguyen The Kiet
5. Student1691060-Huynh Nhat Duy

1	Project Team Information:	3
2	Problem Definition Document:	5
3	Game Project Plan:	6
3.1	Project Details:	6
3.2	Project Initiation / Requirement Documents:	6
3.3	Deliverables:	6
3.4	Project Dependencies:	6
3.5	Major Project Checkpoints:	7
3.6	Quality Plan:	7
4	Game Standard Document:	8
5	User Interface Design Document:	9
5.1	User Interface Creation Process:	9
5.2	Functionality of User Interface and Elements:	10
6	Coding Standards Document	11
6.1	Programming Standards:	11
6.2	Standards for Code Writing Style:	11
6.3	Standards for Declaring Variables:	12
6.4	Standards for Functions Declarations:	12
6.5	Other Standards:	12
7	Task Allocation Document:	13

1 Project Team Information:

Capstone Project Title:	Cyberpunk Detective
Group Members:	Trịnh Tấn Lộc(Leader) Huỳnh Nhật Duy Nguyễn Thế Kiệt Trần Quốc Ân Phạm Thiên Phúc
Problem Statement:	Traditional linear detective games often lack engaging interactive deduction mechanics, leading to passive gameplay. Players need an interactive 3D investigation environment where they can systematically analyze crime scenes, collect clues, solve environmental puzzles, interrogate witnesses, and deduce the correct suspect through structured evidence correlation
Objectives:	Develop a fully functional 3D offline single-player detective game in Unity. Implement a complete investigation case containing: 1 Crime, 1 Location, 1 Victim/Target, 4 Predefined Suspects, 5 Interactive Evidence Items, 2 Logic Puzzles, and Witness Dialogue systems. Build a dynamic Deduction UI (Evidence Notebook) connecting clues to suspects. Provide local JSON data persistence for saving progression and case state.
Capstone project description:	An interactive detective simulation game where the player acts as a lead investigator resolving a murder case. Players explore a 3D crime scene, discover hidden evidence, solve locked-puzzle mechanisms, interview witnesses and suspects, and make a final accusation based on evidence relevance and sequence of events
Capstone project Deliverables:	• Executable Game Build (.exe for Windows PC). • Complete C# Source Code & Unity Project Directory • SRS & Technical Documentation (Form 1 - Form 7).

	• User Manual & Walkthrough Guide • Presentation Slides & Video Demo.
Key milestones:	Day 1: Choose Project, Task Allocation, Complete Problem Definition Day 2: Case Design, Asset Setup, & Environment Layout. Day 3: Player Controller, Object Interaction, & Evidence System. Day 4: Dialogue Manager, Puzzle Mechanics, & Deduction Notebook UI. Day 5: JSON Save/Load System, QA Testing, Polishing, & Documentation.
Constraints:	Strict TechWiz competition time limit (120 Hours) Desktop PC target platform. Local offline execution
Estimated Capstone Duration:	120 Hours/ 5 Days
Notes/Comments	Built using Unity 2022.3.62f3 LTS, C# Scripts, and JSON Data Storage.

Submission Date:	September 29 2026
Student Name:	Trịnh Tấn Lộc
Trainer Name:	Nguyễn Hồng Sơn

2 Problem Definition Document:

Currently, the gaming industry is experiencing rapid growth with a wide variety of titles across different genres. Among them, the detective genre is gradually gaining widespread attention due to the appeal of role-playing, where players can hunt down evidence and become the Sherlock Holmes they have always wanted to be. However, there are still certain limitations in current titles, such as poor evidence data management that complicates the storyline and suspect-tracking quest states, disjointed and inconsistent scripts, and manual programming that leads to event trigger errors and gameplay progression bottlenecks.

Recognizing these issues and aiming to capture this market trend, Nova Nexus Studios is launching the project "Cyberpunk Detective: Decode the Crime"—an offline, single-player detective game set in the high-tech megacity of Neo Zenith in the year 2148. In the game, players take on the role of Detective Kael Orion from the Cyber Crime Investigation Bureau (CCIB) on a mission to solve a high-profile cybercrime. The project is designed to thoroughly solve the challenge of creating a deep, immersive gaming experience that focuses on intellect, observation, forensic clue gathering, and deductive reasoning, rather than conventional action-combat mechanics.

Submission Date:	September 29 2026
Student Name:	Trịnh Tấn Lộc
Trainer Name:	Nguyễn Hồng Sơn

3 Game Project Plan:

3.1 Project Details:

Name of the Client: Aptech / TechWiz Examination Jury

Date of Project Plan: September 29 2026

Project Vision / Objectives: Deliver a fully interactive 3D detective investigation game featuring smooth object exploration, clue gathering, witness interrogation, and logical deduction verification.

Scope: Single investigation case encompassing:

- Crime: Money and Love
- Location: Apartment, Neo Zenith City (Year 2148)
- Victim/Target: Ms. H (36 years old, Single, Center Employee)
- Suspects:
 - + Mr. A (32 years old - Close Colleague & Childhood Friend)
 - + Mr. D (Ex-lover - Has financial disputes & recent heated arguments)
 - + Ms. K (28 years old - Subordinate with Boxing background)
 - + Mr. L (29 years old - Upstairs Neighbor who discovered the body)
- Evidence: Blood-stained Murder Knife, Two-Layer Torn Paper Slip (Cryptic Cipher), Black Sedan Vehicle with Blood Traces, Mismatched Shoes, Unsaved Burner SIM Call Logs
- Puzzles:
 - + Puzzle 1: Paper Slip Assembly (Alpha-Numeric Cipher A=1, B=2... - > Decodes to "ANH A")
 - + Puzzle 2: Safe Unlocking Cipher (Passcode: 0356514251)
- Witnesses:
 - + Mr. L (29 years old - Upstairs Neighbor who discovered the body)
 - + Mr. P (Creditor - Provided debt & affair background information)

Project Organization with Responsibilities:

3.2 Project Initiation / Requirement Documents:

- Case Script & Dialogue Tree Specifications.
- 3D Environment & Furniture Asset Packs.
- Audio Effects (Ambient sound, UI click, Item pickup, Door unlock).

3.3 Deliverables:

- Executable Application (.exe).
- Document Forms 1 through 7.
- User Manual & Walkthrough Guide.
- Source Code Repository (Git/GitHub link).

3.4 Project Dependencies:

- Unity 2022.3.62f3 LTS Engine
- Newtonsoft JSON Library for serialization.

3.5 Major Project Checkpoints:

Checkpoint 1: Project setup, asset importing, camera controller.

Checkpoint 2: Environment setup, collider placement, lighting setup.

Checkpoint 3: Raycast Object Interaction System (Picking up & inspecting clues).

Checkpoint 4: Dialogue & Interrogation System with witness NPCs.

Checkpoint 5: Implementing 2 mini-puzzles (Safe Cipher & Letter Assembly).

Checkpoint 6: Evidence Notebook UI (Clue log, Suspect profiles, Deduction board).

Checkpoint 7: Accusation Logic & Resolution System (Correct/Incorrect suspect triggers).

Checkpoint 8: Local JSON Save/Load System implementation.

Checkpoint 9: Playtesting, Bug fixing, and FPS Optimization.

Checkpoint 10: Final Build creation and documentation sign-off.

3.6 Quality Plan:

- **Review Activities:** Daily 15-minute scrum meetings to check code integration and case logic consistency.
- **Testing Activities:** Unit testing on dialogue node switching and item inventory state, integration testing on puzzle unlock triggers and deduction board validation, end-to-end playtesting to ensure case resolution triggers correctly upon selecting the true suspect.

Submission Date:	September 29 2026
Student Name:	Trịnh Tấn Lộc
Trainer Name:	Nguyễn Hồng Sơn

4 Game Standard Document:

Property	Value
Playtesting	Conducted internally by QA members and peer-reviewed to ensure clue progression is logical and bug-free.
Level design effectiveness	Crime scene map designed with distinct investigation areas (Study, Safe Room, Hallway) to naturally guide player flow.
UI of the Game	Clean, diegetic Detective Notebook UI with tabs for Evidence, Suspects, Puzzles, and Final Accusation
Player Graphics and Dynamics	First-person smooth camera movement, interactive object inspection in 3D space with rotate/zoom controls.
Game difficulty and balance	Balanced puzzle difficulty with contextual hints available in the Evidence Notebook if the player gets stuck.
Player engagement and immersion	Suspenseful Cyberpunk-style ambient sound, interactive clue discovery, and dramatic witness dialogue choices
Controls responsiveness and intuitiveness	Intuitive WASD movement, Mouse Aim, 'E' to Interact, 'Tab' to open Notebook, Esc for Pause Menu.
Overall performance and satisfaction of the game	Maintained steady 60 FPS performance on target PC hardware with zero critical script exceptions.

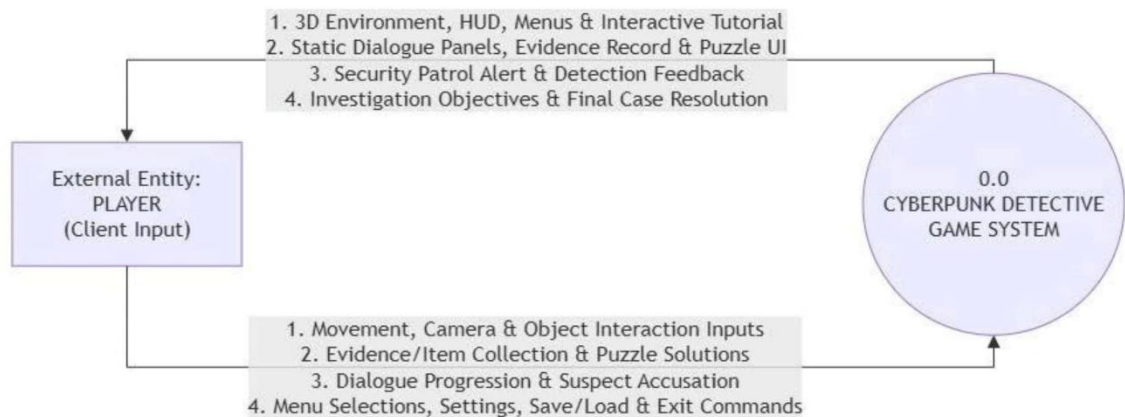
Submission Date:	September 29 2026
Student Name:	Trịnh Tấn Lộc
Trainer Name:	Nguyễn Hồng Sơn

5 User Interface Design Document:

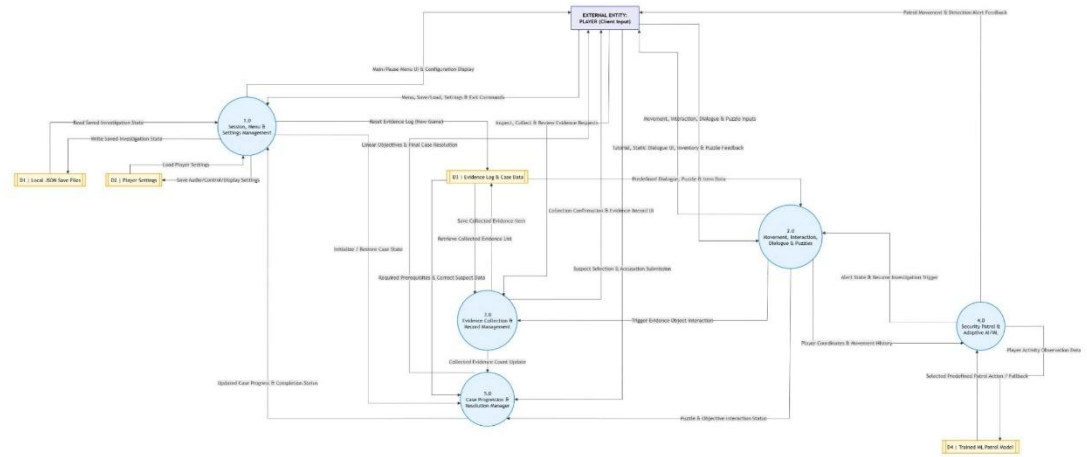
5.1 User Interface Creation Process:

- **Wireframing:** Low-fidelity mockups created for Main Menu, In-game HUD, Detective Notebook, Dialogue Overlay, and Accusation Screen.
- **Asset Styling:** Styled UI components using a noir brown/leather paper texture for the Detective Notebook and minimalist white/gold text for HUD popups.
- **Unity Canvas Setup:** Implementation using Unity UI Canvas with Scaler set to Scale With Screen Size (1920x1080 reference resolution).
- **Event Binding:** C# Event Listeners hooked to UI buttons, dialogue option triggers, item inspection popups, and clue unlock notifications.

DFD Level 0



DFD Level 1



5.2 Functionality of User Interface and Elements:

UI Component	Elements	Functionality
Main Menu	New Game, Load Game, Settings, Quit	Starts investigation, loads JSON save file, adjusts resolution/volume, exits app
Investigation HUD	Reticle, Interaction Prompt ("Press E"), Notification Banner	Displays crosshair, shows interactable object names, displays "New Evidence Added" alerts.
3D Item Inspection View	3D Model Viewer, Rotate/Zoom Controls, Description Panel	Renders selected evidence in 3D overlay, allows player to rotate item to find hidden marks.
Detective Notebook (Tab Key)	Evidence Tab, Suspects Tab, Sequence Tab, Accuse Button	Displays 5 collected clues, shows 4 suspect profiles, allows linking clues to statements.
Dialogue Overlay	Speaker Portrait, Nameplate, Dialogue Text, Choice Buttons	Displays NPC speech, allows player to select interrogation questions or present evidence
Puzzle UI Overlay	Interactive Cipher Dial / Jigsaw Pieces, Reset	Modal window for solving puzzle; updates inventory upon solution

	Button, Exit	
Final Resolution Screen	Suspect Selection List, Confirm Accusation, Verdict Video/Text	Player selects final culprit; displays success sequence if correct suspect chosen, or failure outcome if wrong

Submission Date:	September 29 2026
Student Name:	Trịnh Tấn Lộc
Trainer Name:	Nguyễn Hồng Sơn

6 Coding Standards Document

6.1 Programming Standards:

- Language: C# using standard .NET Framework guidelines in Unity 2021.3 LTS.
- Architecture: Component-Based Architecture with Manager Pattern (CaseManager, InventoryManager, DialogueManager, SaveSystem).
- Object-Oriented Principles: Encapsulation for private fields, Interfaces (IInteractable) for interactive world items.

6.2 Standards for Code Writing Style:

- Class Names: PascalCase (e.g., EvidenceItem, SuspectProfileManager, CipherPuzzleController).
- Method Names: PascalCase with verbs describing action (e.g., CollectEvidence(), VerifyAccusation(), SaveGameState()).
- Variable Names:
- Private fields: camelCase with underscore prefix (e.g., _evidenceCount, _isPuzzleSolved).
- Public / Serialized fields: camelCase (e.g., interactDistance, suspectName).
- Constants: ALL_CAPS_WITH_UNDERSCORES (e.g., MAX_EVIDENCE_COUNT = 5).
- Indentation: 4 spaces (standard C# tab formatting), brace placement on new line (Allman style).

6.3 Standards for Declaring Variables:

- Always specify explicit data types instead of var unless type is obvious from constructor.
- Use [SerializeField] private instead of public fields to expose variables to Unity Inspector while maintaining encapsulation.
- Group variables logically using [Header("...")] and [Tooltip("...")] attributes in Unity scripts.

6.4 Standards for Functions Declarations:

- Single Responsibility Principle: Each function must perform exactly one logical operation.
- Functions should not exceed 40 lines of code; break complex calculations into helper methods.
- XML documentation comments (/// <summary>) required for all public methods and manager classes.

6.5 Other Standards:

- Version Control: Git repository with strict .gitignore for Unity temporary files (Library/, Temp/, Logs/).
- Data Serialization: Use JSON (JsonUtility or Newtonsoft.Json) for storing player progression, unlocked evidence flags, and puzzle state locally.
- Null Safety: Always check null or use null-conditional operators (?.) before calling component references to prevent NullReferenceException at runtime.

Submission Date:	September 29 2026
Student Name:	Trịnh Tấn Lộc
Trainer Name:	Nguyễn Hồng Sơn

7 Task Allocation Document:

https://docs.google.com/spreadsheets/d/1ghEXUIGZbcpp6ueTtOOYbr4N9AAIYPOuBFShSwd_Ftc/edit?usp=drive_link

